Prototype Design Document

Group: *Remote, Passing Grade+*

Members: *Ernest Khaianen, Roman Kuzero, Aapo Niklander*

The overall use-case

An app for fans of the Finnish ice hockey league, Liiga. The user can pick their favourite team, and the app displays their upcoming matches for an amount of time chosen by the user. The app displays the following information about the upcoming matches: place, date and time, weather forecast for chosen place and time, weather camera image, and ticket prices.

Data sources for this app are

- Liiga's API for data relating to the teams (https://liiga.fi/api/v2/)
- FMI's API for the weather forecast (https://en.ilmatieteenlaitos.fi/open-data)
- Digitraffic's API for the weather camera images (https://www.digitraffic.fi/en/)
- Lippu.fi has data but there is no ready API so some workarounds are due

Liiga's API is not publicly documented, but the API and corresponding data are accessible for free. Aapo has a pre-existing Python script for fetching data from Liiga's API, but handling the other API's, data visualization, user interface etc. will need to be implemented from scratch.

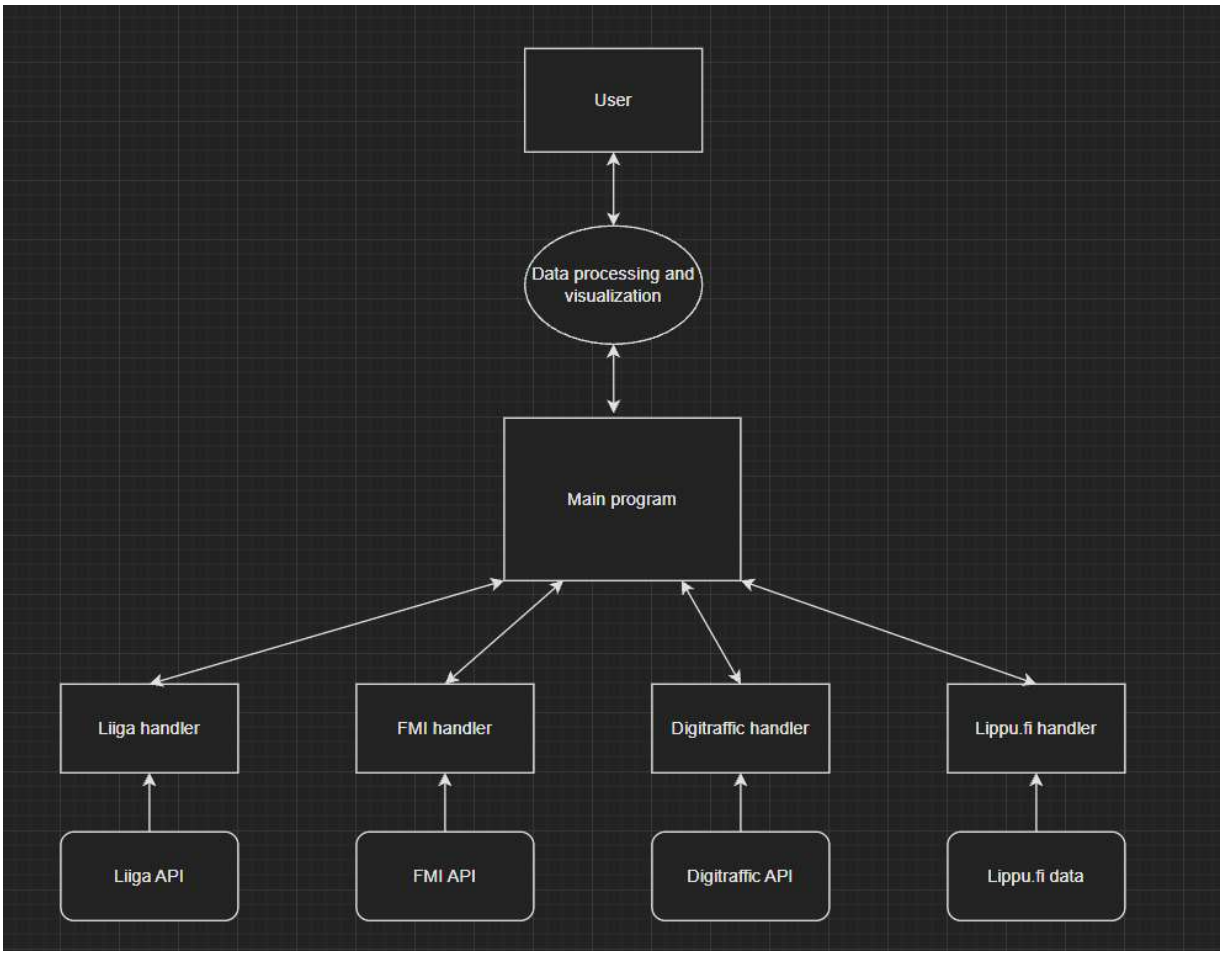
Planned technologies

We are planning to use Python for API usage and handling the data. Godot engine is used for prototype GUI design. The user interface will be implemented as a web application / a desktop application made with Python?. it's apparently allowed to make a web app as long as the backend is written in C++/java/python

Planned AI usage going forward? Moderate usage of AI for testing and research might prove useful, but as a main rule agreed in our team there would be no straight copy-paste usage of AI-generated code.

Main components of the application

The components for the application are a main program for data visualization and graphics/UI, and separate components for each API. When the user selects a team, the main component calls to the other components to fetch current data from the APIs. The data is then handled by the main component and presented to the user filtered by a temporal parameter set by said user. In case some data was not reachable or existing (like weather forecasts or traffic camera pictures) the absent data is not shown.



Description of how the components of the program interact with each other.

Division of labour

Our group originally had four members, but Viljami didn't ~~make it~~ respond when we tried to contact him regarding the project. We then decided to move forward with three members. The prototype UI sketch was mainly made by Ernest, and the design document was composed by Roman and Aapo. All the parts were then verified by all the members of the group. In the future the division of labour is unsure, but tentatively Ernest will focus on graphical design, and Roman and Aapo focus on fetching and handling the data from the different APIs. This is to be subject to change through the implementation of the project.

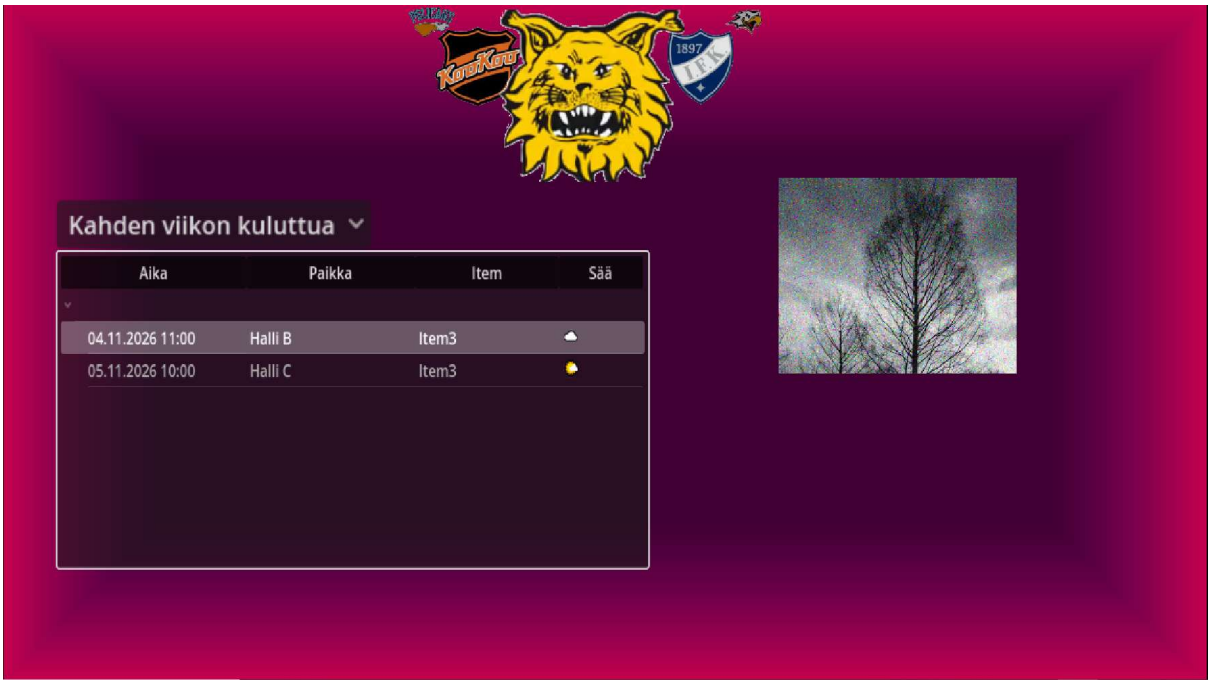
Version control

The version control is provided by using git-lab.

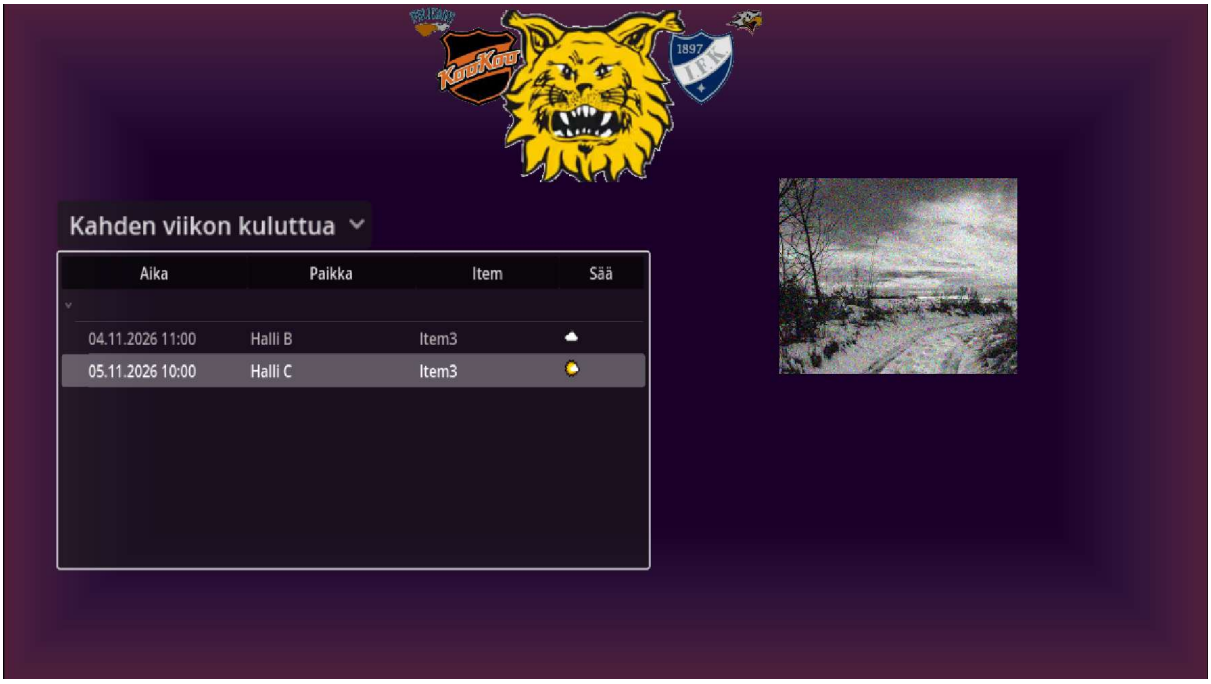
Testing

Appropriate unit-tests and their implementation will be decided further into the project.

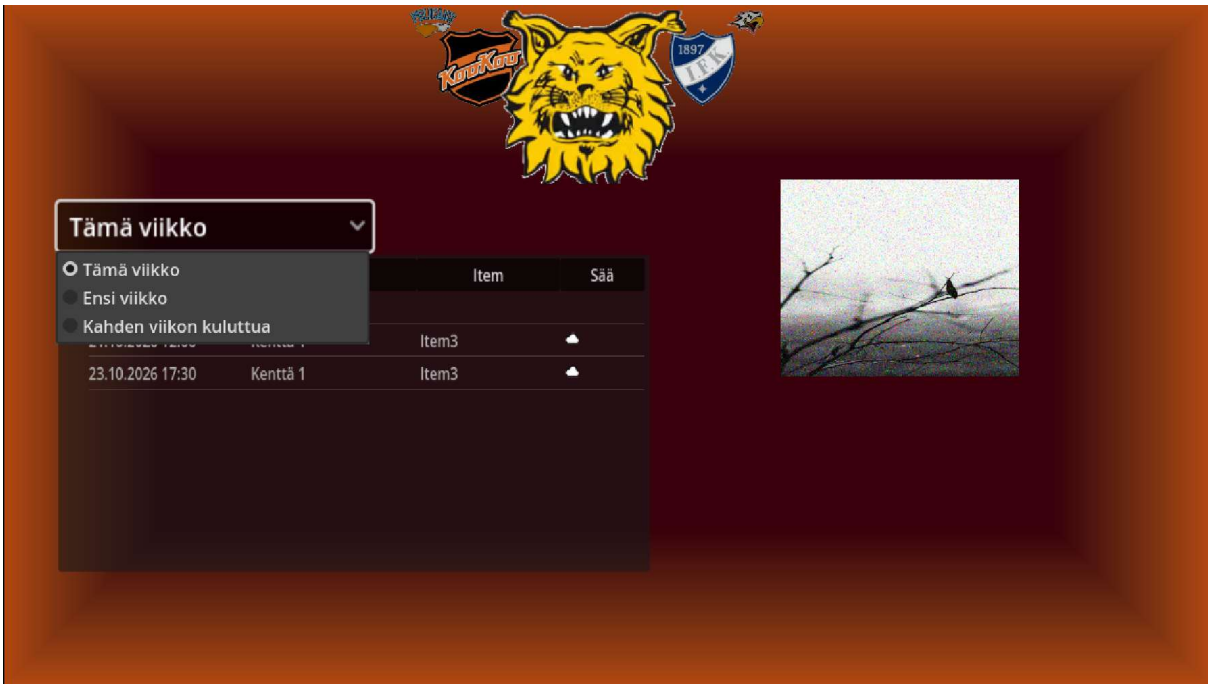
GUI Prototype



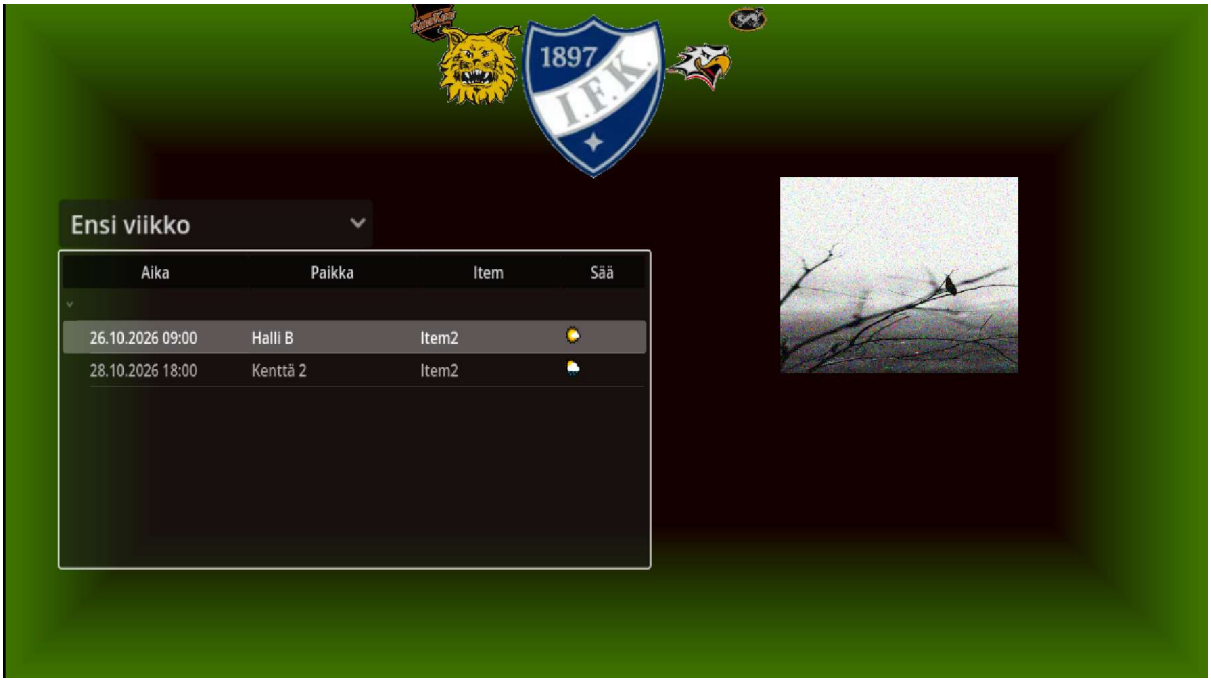
The main view of the application prototype showing LYNX -team's events coming in two weeks, time, date, place, the weather and a picture from a local surveillance camera for the chosen event. Note that the picture provided by the Digitraffic's API shows what it looks like at the moment of requesting the picture, not in the prospect. The logos and the info on the events is provided by the Liiga's API. The emoji representing weather is based on the forecast by FMI's API.



The picture changes when the user chooses another event.



The drop-down menu allows the user to choose the time period to filter the events.



A carousel of logos allows users to select the preferred team to browse the events the team participates into.